Vansh Varshney

B.Tech Student · Machine Learning & Applied Research
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RESEARCH STATEMENT

Second-year undergraduate in AI & Data Science with a focused interest in the foundations of machine learning — particularly how models generalise, fail, and can be made more efficient. Self-directed study of gradient-based optimisation, probabilistic reasoning, and neural network theory (via Karpathy's neural net series and 3Blue1Brown's linear algebra and calculus curriculum) complements formal coursework. Seeking to contribute to exploratory research at Google — to move from applying ML techniques to understanding and questioning them.

EDUCATION

B.Tech — Artificial Intelligence & Data Science CGPA: 8.3 / 10
GGSIPU (USAR), New Delhi | 2024 – 2028

Class XII — CBSE 2024 88.2%

RESEARCH INTERESTS

Machine Learning theory (generalisation & optimisation) • Efficient ML systems (quantisation, inference optimisation) • Representation learning • Data-centric AI • Applied statistics

TECHNICAL SKILLS

ML & Theory:	Supervised & Unsupervised Learning, Bias-Variance, Optimisation, Probabilistic Models
Mathematics:	Linear Algebra, Probability & Statistics, Calculus — applied to ML foundations
Programming:	Python (Pandas, NumPy, Scikit-learn), SQL, C, Java — Jupyter Notebook, Git
Visualisation:	Matplotlib, Seaborn — communicating experimental results and data distributions
Concepts:	Model Evaluation (cross-validation, ROC-AUC), Feature Engineering, EDA, ETL

PROJECTS & INDEPENDENT RESEARCH

LLM Inference Efficiency — DeepSeek R1 8B
Systems ML · Quantisation · Empirical Study

– Investigated how 4-bit quantisation affects inference quality and memory footprint of a large language model deployed on a 6 GB VRAM GPU — motivated by the question of how much precision a model truly needs.
– Empirically measured throughput and memory trade-offs across quantisation settings; achieved ~35% speed improvement within fixed hardware constraints, with minimal degradation in output quality.
– This project shaped an ongoing interest in efficient ML systems and the gap between research-scale and deployment-scale models.

Generalisation Study — ML Model Comparison
Model Theory · Bias-Variance · Empirical ML

– Posed a research question: under what conditions do simpler models outperform complex ones on structured tabular data?
– Benchmarked models of varying complexity using learning curves, cross-validation, and held-out evaluation; documented cases where regularised linear models matched or exceeded tree ensembles.
– Developed practical intuition for overfitting, underfitting, and the role of dataset size in model selection — grounding theoretical study in observable experimental outcomes.

Churn Prediction — Feature & Model Analysis
Classification · Statistical Reasoning

– Framed customer churn as a binary classification problem; systematically compared algorithms on precision, recall, and ROC-AUC, treating model selection as a hypothesis-driven process.
– Applied feature importance analysis to identify the strongest behavioural predictors; findings revealed that recency of engagement dominated frequency — a non-obvious result warranting further study.

Independent Study — Neural Networks & Backpropagation
Self-Directed · Theory-Driven Learning

– Worked through Andrej Karpathy's Micrograd and neural network lecture series, implementing backpropagation from scratch to build mechanistic understanding beyond library-level abstraction.
– Studied 3Blue1Brown's Essence of Linear Algebra and calculus series to strengthen the mathematical intuition underlying gradient descent and loss landscapes.

CERTIFICATIONS

– IBM SkillBuild AI Track • Google Cloud Vertex AI • Generative AI with Gemini (Google)